HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Actuarial Mathematics and Statistics

F79SU

SURVIVAL MODELS

IN-PERSON EXAMINATION

April Examination and Assessment Diet 2024

Duration: 2 hours

Total marks available: 60

Answer all 6 questions

Attached: No additional documents attached

Supplied: Formulae and Tables for Examinations of the Institute and Faculty of Actuaries

Calculator Permitted: Yes

Ethical Commitment: By submitting answers to this paper, I declare that these answers are entirely my own work and have not been shared, in part or in whole or in any form, with any other student, or disseminated in any way.

I also confirm that I have reviewed and understand all University regulations, policies and procedures on assessment misconduct, including but not limited to accessing unauthorised materials during this examination or the use of unauthorised electronic devices.

I understand that infringing this statement would represent an academic misconduct offence subject to disciplinary action according to the University Regulations and Policy regarding Academic Misconduct, with possible significant consequences for degree progression and final degree outcome.

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1. A portfolio of insured lives is classified into three groups by educational attainment: (A) people who never attended university, (B) people who attended a university but never got a Bachelor's degree, and (C) people who obtained a Bachelor's or higher qualification.

The force of mortality at age t of a life is denoted by $\lambda(t;\mathbf{z})$ where $\mathbf{z}$ is associated with different levels of educational attainment such that $\mathbf{z} = (z_B, z_C)$. Suppose that $\lambda(t;\mathbf{z})$ follows a Cox proportional hazards model as

$$\lambda(t;\mathbf{z}) = \lambda_0(t)\exp\left(\beta_B z_B + \beta_C z_C\right),$$

where $\boldsymbol{\beta} = (\beta_B, \beta_C)$ represents the coefficients associated with different levels of educational attainment.

Assume that the group of people who never attended university (A) is chosen to be the reference level, let

$$z_B = \begin{cases} 1 & \text{if a life is in group B} \\ 0 & \text{otherwise.} \end{cases}$$

and suppose that z_C is defined in a similar manner.

(a) Write down expressions for $\lambda_A(t)$, $\lambda_B(t)$ and $\lambda_C(t)$ for the three levels of educational attainment (A, B and C) in terms of a baseline hazard $\lambda_0(t)$ and the related regression coefficients.

[3 marks]

(b) Suppose now that the baseline hazard $\lambda_0(t)$ follows the Gompertz law of mortality, that is $\lambda_0(t) = \exp(a+bt)$. The model is fitted to data and the following estimates are obtained: $\hat{a} = -5$, $\hat{b} = 0.01$, $\hat{\beta}_B = -0.85$ and $\hat{\beta}_C = -1$.

(i) Estimate the force of mortality for a life in group A aged 65.

(ii) Estimate the force of mortality for a life in group B aged 65.

(iii) Estimate the force of mortality for a life in group C aged 65.

[3 marks]

(c) Plot the log of the force of mortality of groups A and C (on the same graph) against t, for $65 < t < 75$. Your graph does not need to be precise, but it should show the main trends and differences for the two groups.

[2 marks]

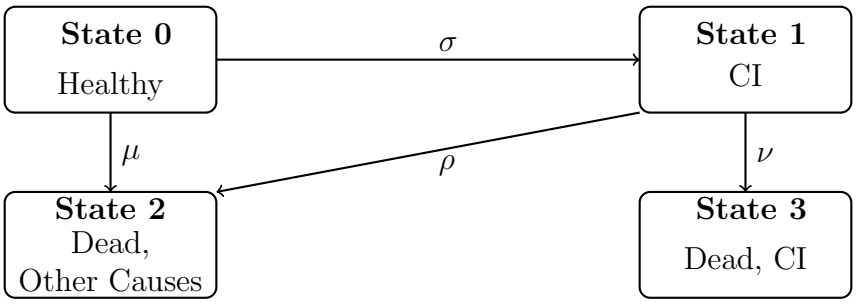
(d) Comment on the impact of education on mortality for groups A and C, based on your graph from part (c).

[1 marks]

[Total 9 Marks]

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2. The following 4-state Markov model with constant transition intensities, σ, μ, ρ, ν, is used to model a critical illness (CI) insurance portfolio for a group of insured people aged x. The model describes the life history of an initially 'healthy' insured individual (State 0) based on two events: incidence of CI (State 1), and death from other causes (State 2) or death from CI conditions (State 3).



(a) Write down expressions for ${}_tp_x^{00}$ and ${}_tp_x^{11}$. [2 marks]

(b) Suppose that the observation on the ith life is

v_i = total waiting time in State 0
w_i = total waiting time in State 1
s_i = 1 if a healthy individual is diagnosed with a CI condition, 0 otherwise
d_i = 1 if a healthy individual dies from other causes, 0 otherwise
u_i = 1 if an individual with CI dies from other causes, 0 otherwise
r_i = 1 if an individual with CI dies from CI, 0 otherwise.

(i) Write an expression for the likelihood function considering n independent lives.

[2 marks]

(ii) Find estimates $\hat{\sigma}$, $\hat{\mu}$, $\hat{\rho}$, $\hat{\nu}$ of σ, μ, ρ, ν, respectively, based on the data collected from a large number of individuals with various ages:

200: Number of healthy individuals that become critically ill.
100: Number of individuals that die from CI.
300: Number of individuals in State 0 that die from other causes.
30: Number of individuals in State 1 that die from other causes.
8000 years: Total time in State 0 amongst all individuals.
1500 years: Total time in State 1 amongst all individuals.

[3 marks]

(iii) By calculating approximate 95% confidence intervals for $\hat{\mu}$, $\hat{\rho}$ and $\hat{\nu}$, comment on the above model.

[4 marks]

[Total 11 Marks]

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3. Suppose you have collected the following mortality data for 4 lives with ages between 50 and 51 where

- a_i is the time after age 50 when the ith life came under observation,
- b_i is the time after age 50 when the ith life may be censored,
- t_i is the time after age 50 when the observation for the ith life actually ended.

For this question, assume that all uncensored observations were deaths. The data are as follows:

Life	a_i	b_i	t_i
John	0.25	0.75	0.6
Timothy	0.8	1	1
Ashley	0	0.8	0.8
Britney	0.5	0.75	0.7

(a) Work under the binomial model of mortality and assume constant force of mortality between ages 50 and 51. Let us denote this force of mortality as μ. Find an expression for the the likelihood function for μ. [3 marks]

(b) Verify that $\widehat{\mu} = 1.4966$ is the maximum likelihood estimate of μ. [3 marks]

(c) Let q_{50} be the probability of death between ages 50 and 51, and denote $\widehat{q}_{50}$ to be the maximum likelihood estimate for q_{50}. Compute $\widehat{q}_{50}$. [1 mark]

(d) Consider the hypotheses testing problem $H_0 : q_{50} = 0.5$ versus $H_1 : q_{50} \neq 0.5$. Compute the Wald test statistic. Here you can use the fact that the variance of $\widehat{\mu}$ is 0.4564. [2 marks]

(e) Use significance level 0.05. What is your conclusion in view of the Wald test above? [1 mark]

[Total 10 Marks]

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4. The table below gives the central exposed to risk, E_x^c, the number of deaths, d_x, and the maximum likelihood (ML) estimates $\hat{\mu}_x$ of $\mu_{x+\frac{1}{2}}$ for a set of assured lives, $\mathcal{S}$, for ages $x = 60, \ldots, 70$.

Age x	E_x^c	d_x	$\hat{\mu}_x$	z_x
60	7236	97	0.0134	-0.292
61	6312	95	0.0151	-0.147
62	5762	102	0.0177	0.461
63	5189	101	0.0195	0.395
64	4492	91	0.0203	-0.208
65	2585	58	0.0224	-0.159
66	1585	42	0.0265	0.285
67	1291	32	0.0248	-0.715
68	1120	36	0.0321	0.207
69	962	32	0.0333	-0.183
70	808	32	0.0396	0.230

It is found that $\sum_x z_x = -0.126$ and that $\sum_x z_x^2 = 1.265652$.

(a) A plot of the ML estimates, $\hat{\mu}_x$, against age is given in Figure 1. State briefly (giving reasons) what you would do if you wished to estimate the force of mortality for the purpose of calculating premium rates. [2 marks]

(b) A graduation of the ML estimates using a $GM_{\boldsymbol{\alpha}}^{0,2}(x)$ Poisson generalised linear model, gave the fitted values as $\log(\dot{\mu}_x) = -10.3595 + 0.1013x$. The table above also shows the standardised residuals, z_x, from this model.

(i) A plot of $\dot{\mu}_x$ against age is given in Figure 2. Comment on the suitability of the model. [1 mark]

(ii) Perform an appropriate test to check the overall fit of the graduation. [3 marks]

(iii) Perform the grouping of signs (Stevens) test. What is this test checking for? [3 marks]

(iv) Calculate an approximate 95% confidence interval for μ_{70} based on $\hat{\mu}_{70}$. Hence, comment on the difference between $\hat{\mu}_{70}$ and the graduated estimate $\dot{\mu}_{70}$. [5 marks]

[Total 14 Marks]

Figure 1: ML estimates, $\hat{\mu}_x$, against age, x.

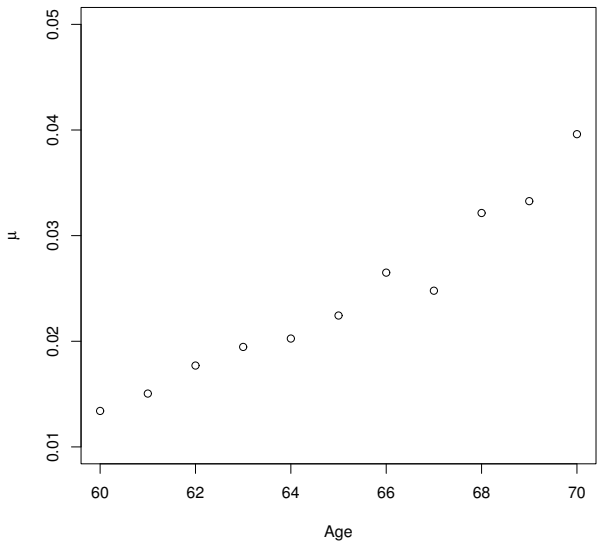
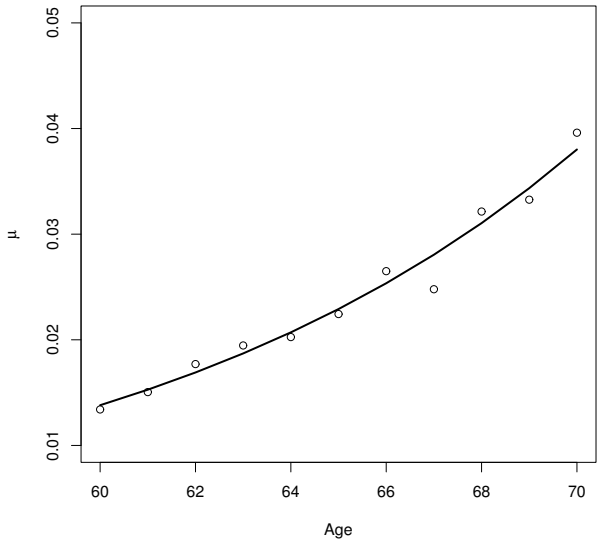


Figure 2: ML estimates, $\hat{\mu}_x$ (circles), and graduated values, $\dot{\mu}_x$ (solid line), against age, x



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5. A Lee-Carter model has been used to forecast mortality in a country, using past data from 1990 to 2023 and projecting in subsequent years from 2024 onwards. The model has the following form:

$$\log(\mu_{x+1/2,y+1/2}) = \alpha_x + \beta_x \kappa_y,$$

for $x = 50, 51, \ldots, 80, \quad y = 1990, 1991, \ldots, 2023$, with future mortality projected using

$$\kappa_{y+1} - \kappa_y = \delta + \epsilon_y,$$

where δ is a trend term and $\epsilon_y \sim N(0, \sigma^2)$ are independent error terms.

Fitting the model has resulted in the following estimates:

$$\hat{\alpha}_{70} = -3.9, \;\; \hat{\beta}_{70} = 0.018, \;\; \hat{\kappa}_{2023} = -29.3, \;\; \hat{\delta} = -0.543, \;\; \hat{\sigma}^2 = 0.892.$$

(a) Calculate the projected value $\hat{\kappa}_{2025}$ and the corresponding 99% prediction interval, based on the estimates given above.

[4 marks]

(b) Calculate the projected log mortality rate $\log(\mu_{70.5,2025.5})$, based on the estimates given above.

[1 mark]

(c) Consider using the model above for obtaining mortality forecasts for the year 2045. Without performing any calculations, comment on the width of a prediction interval for $\hat{\kappa}_{2045}$, as compared to that of $\hat{\kappa}_{2025}$, and the related uncertainty.

[2 marks]

(d) It is suggested that a different model may be used in the above mortality forecasting, having the following form:

$$\log(\mu_{x+1/2,y+1/2}) = \alpha^* x + \beta^*_x \kappa^*_y,$$

with the remaining parts of the model being the same as before and for the same range of x and y.

Comment on why and under what circumstances this model may be preferable.

Suggest how you could compare the fit of this model to that of the model used earlier.

[3 marks]

[Total 10 Marks]

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6. Suppose that in a mortality study lifetimes are not observed directly; instead census data are available on policyholders at risk on 1 July of each year. The following definitions of age for census and death data are used:

$P^C_{x,t}$: number of lives aged x next birthday at time t, where t is the census date;
d_x: number of deaths aged x last birthday on 1 July before death.

Part of the data is given in the table below.

	Census data		
Age	1/7/17	1/7/18	1/7/19
70	6517	6764	6869
71	6878	6964	7192
72	6907	6935	7277

(a) Determine the type and range of the rate interval. [1 mark]

(b) Find from the data in the table the number of lives aged 70 at 1/7/2017, according to the definition of age for deaths, justifying your answer. When will the age label for these lives change to 71? [3 marks]

(c) Suppose that we obtain estimates for μ_{70} and q_{70} based on the definition of age for deaths. State, giving reasons, the exact age to which these estimates apply. Any assumptions made should be clearly stated. [2 marks]

[Total 6 Marks]

[Grand total: 60 marks]

END OF PAPER